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ĐẠI HỌC BÁCH KHOA HÀ NỘI

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

GENERATIVE AI IN BIOINFORMATICS, MEDICINE AND HEALTHCARE

Chapter 6: Explainable AI

ONE LOVE. ONE FUTURE.

Content

6.1. Introduction to Explainable AI

6.2. Explainable AI in GenAI

6.3. Several interpretation methods can be applied to Gen AI

6.4. Evaluate interpretation by incorporating experimental data

6.5 Identify key biological factors in AI models

6.6. Interpretation in modern problems

Do we really mimic Human Mind?

The Human Mind

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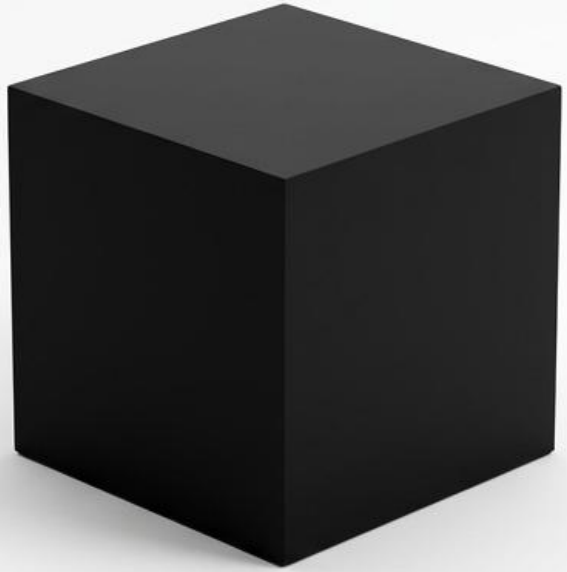
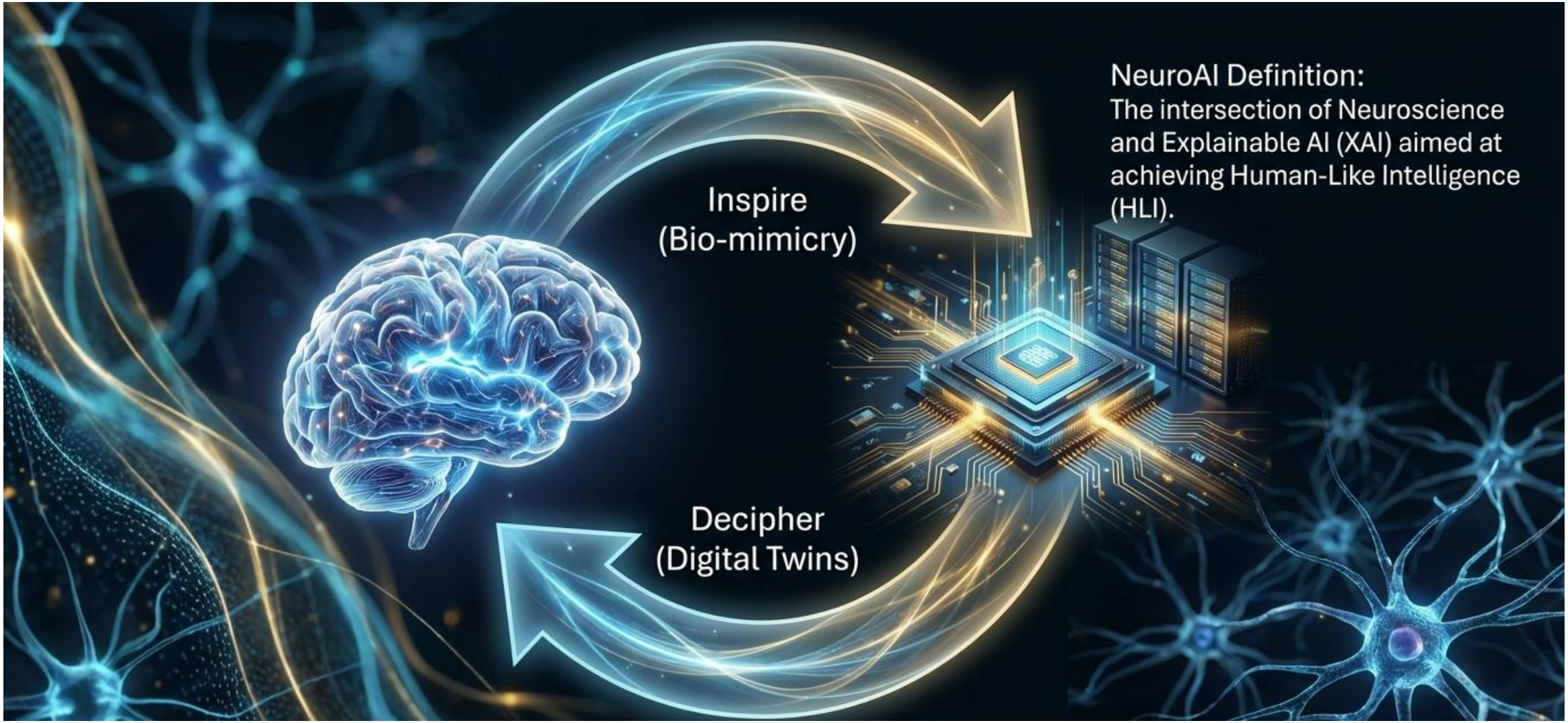
Deep Neural Networks



- **The Shared Challenge:** Processing information through millions of subunits to create adaptive behavior.
- **The Epistemic Gap:** We build systems we do not fully understand, mirroring our own biology.
- **Goal:** Applying Sherringtonian (micro) and Hopfieldian (macro) toolkits to AI transparency.

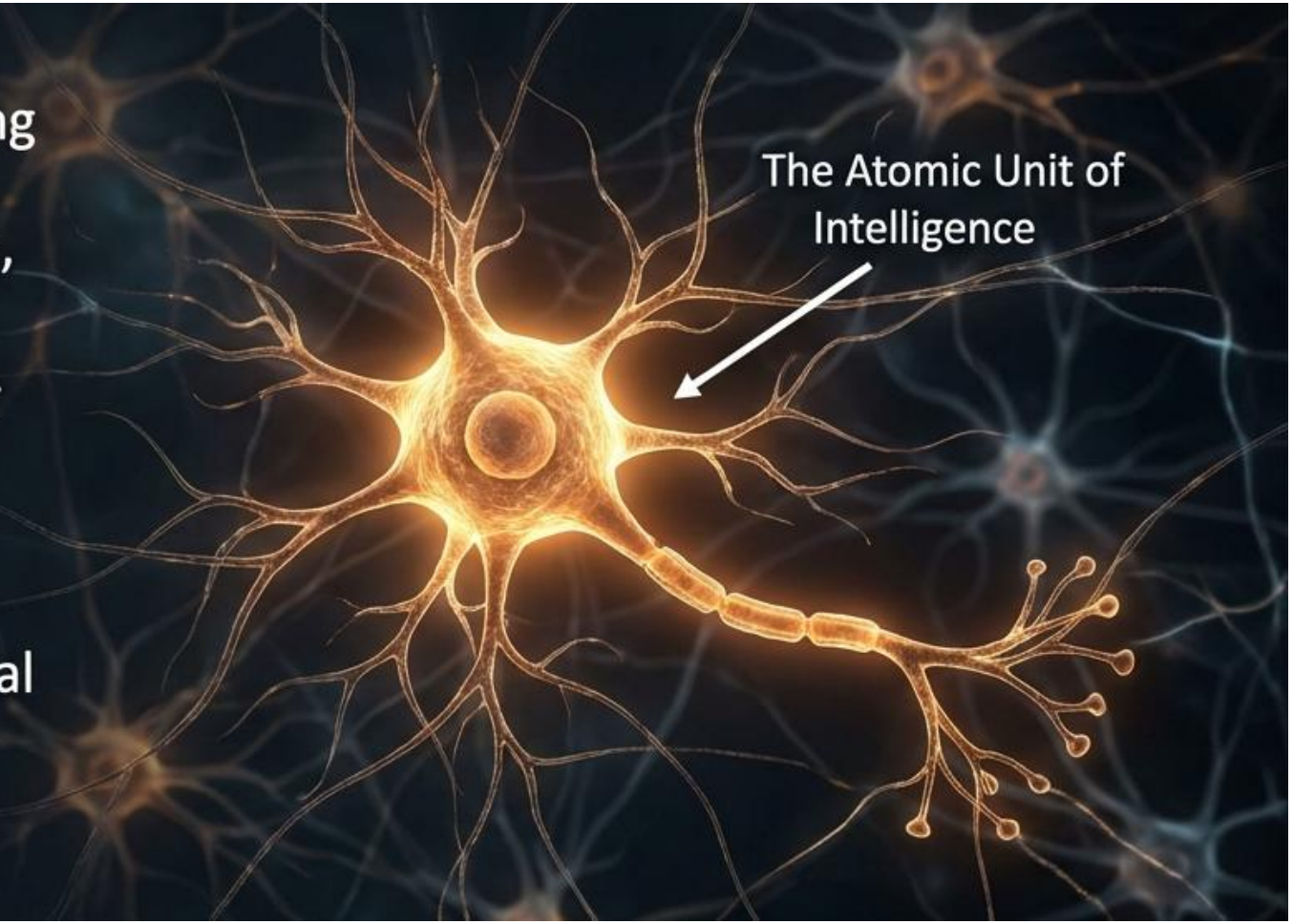
The mind is a black box, and we've studied it for 150 years without opening it.

The circle of learning: reciprocal insight

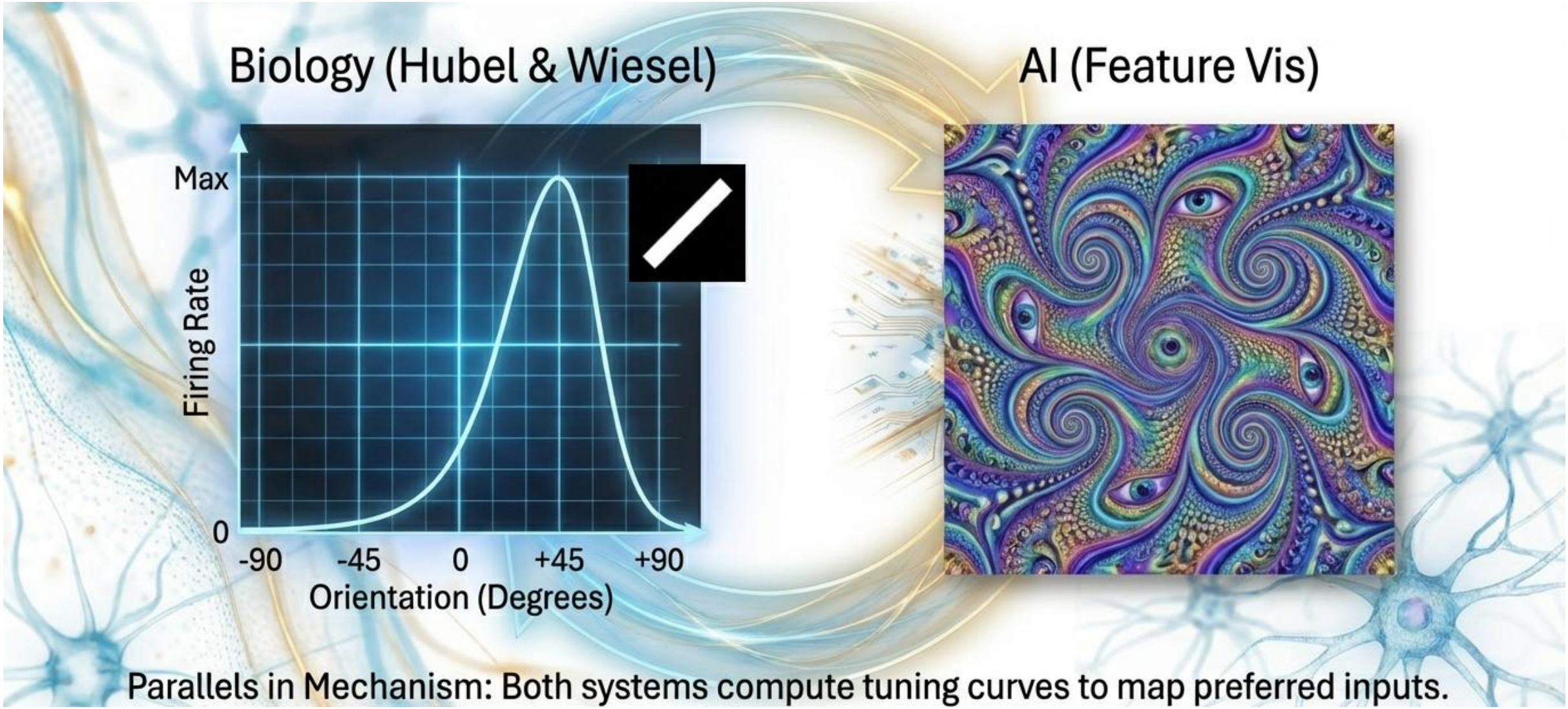


The Micro View

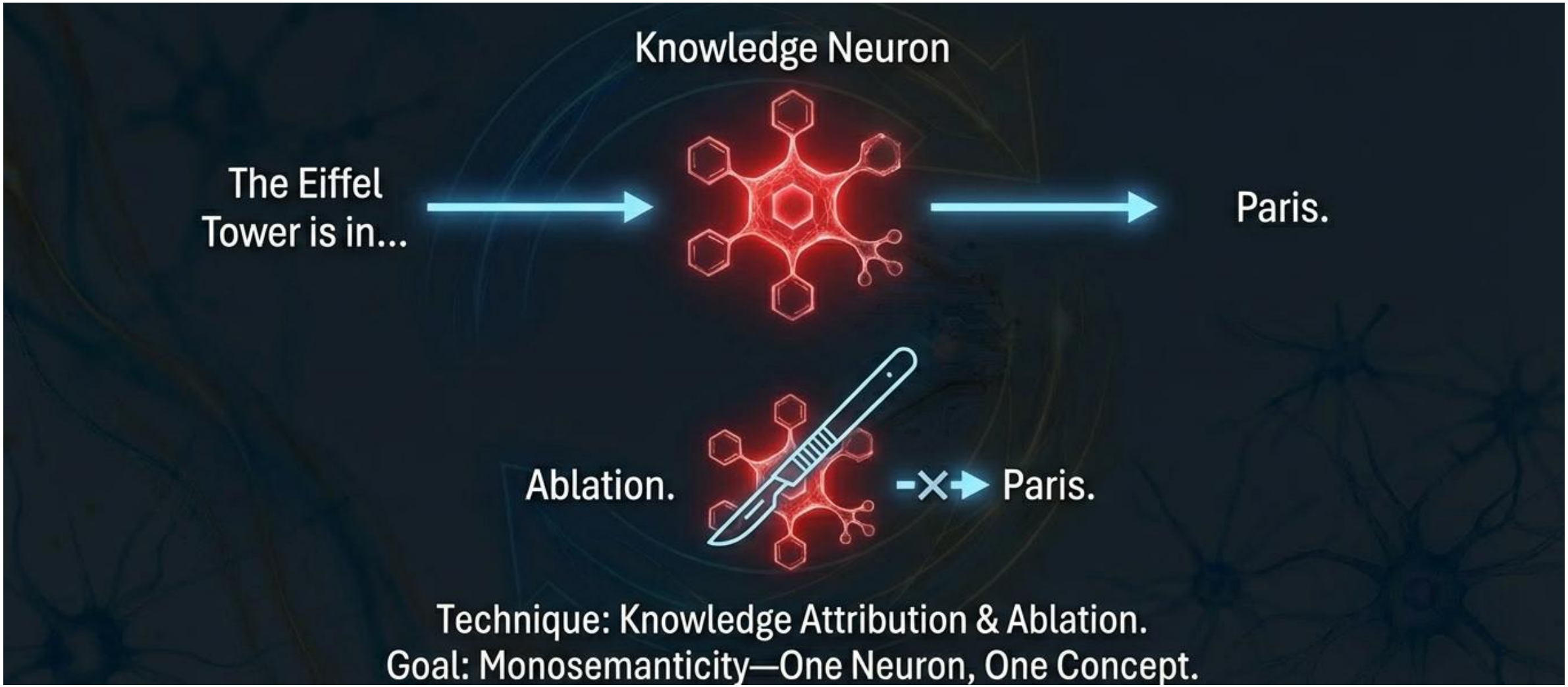
- Sherringtonian View: Understanding Understanding intelligence by characterizing the activity of single, specific neurons.
- Historical Context: Named after Sir Charles Sherrington. Focuses on “labeled lines” of communication.
- AI Relevance: The basis for understanding Convolutional Neural Networks (CNNs) and feature detectors.



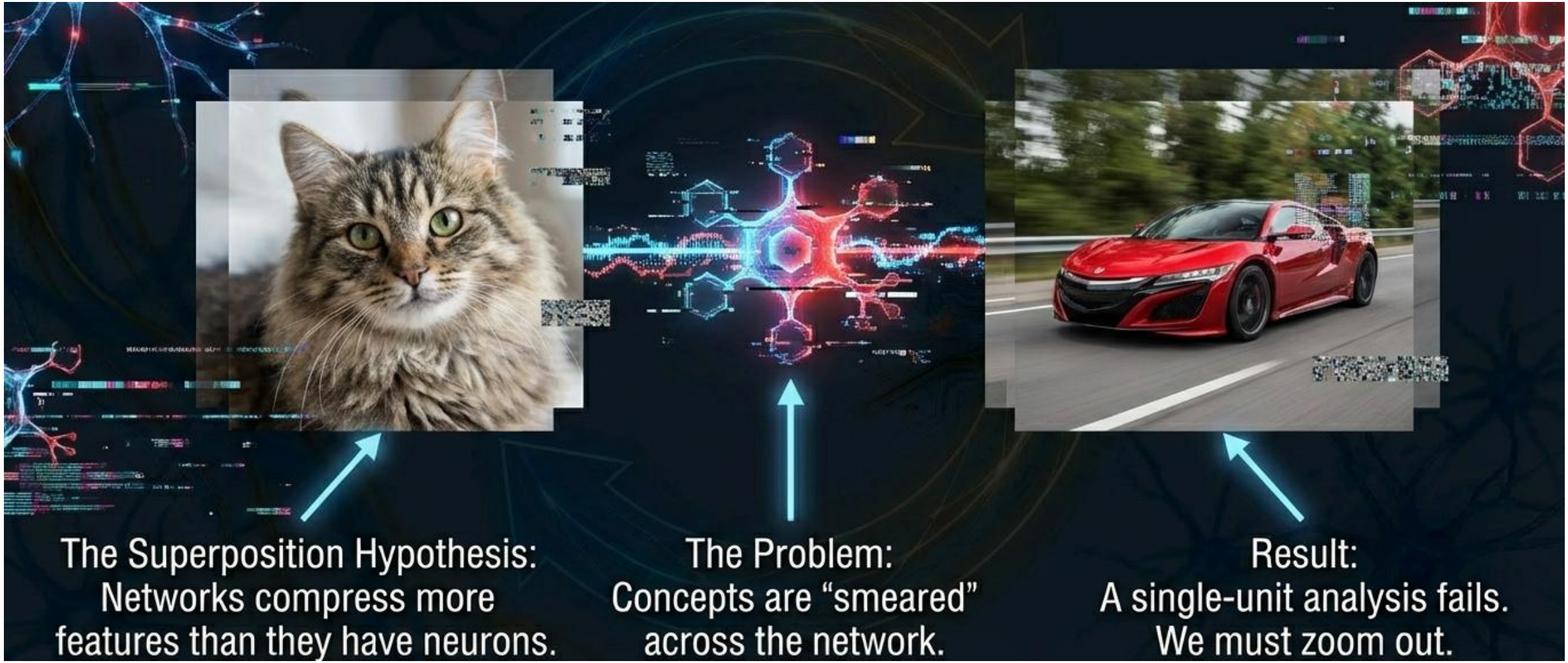
The Receptive Fields vs Feature Visualization



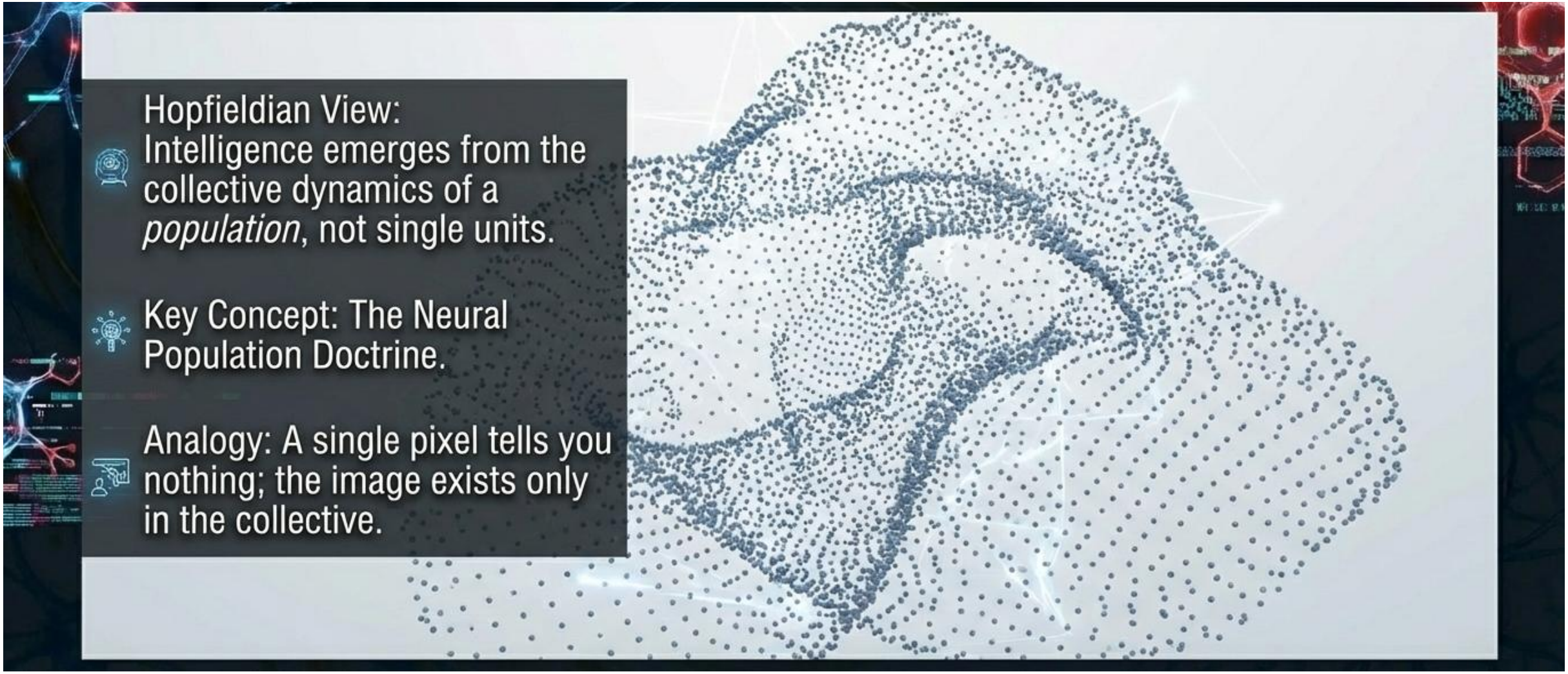
Knowledge Neuron?



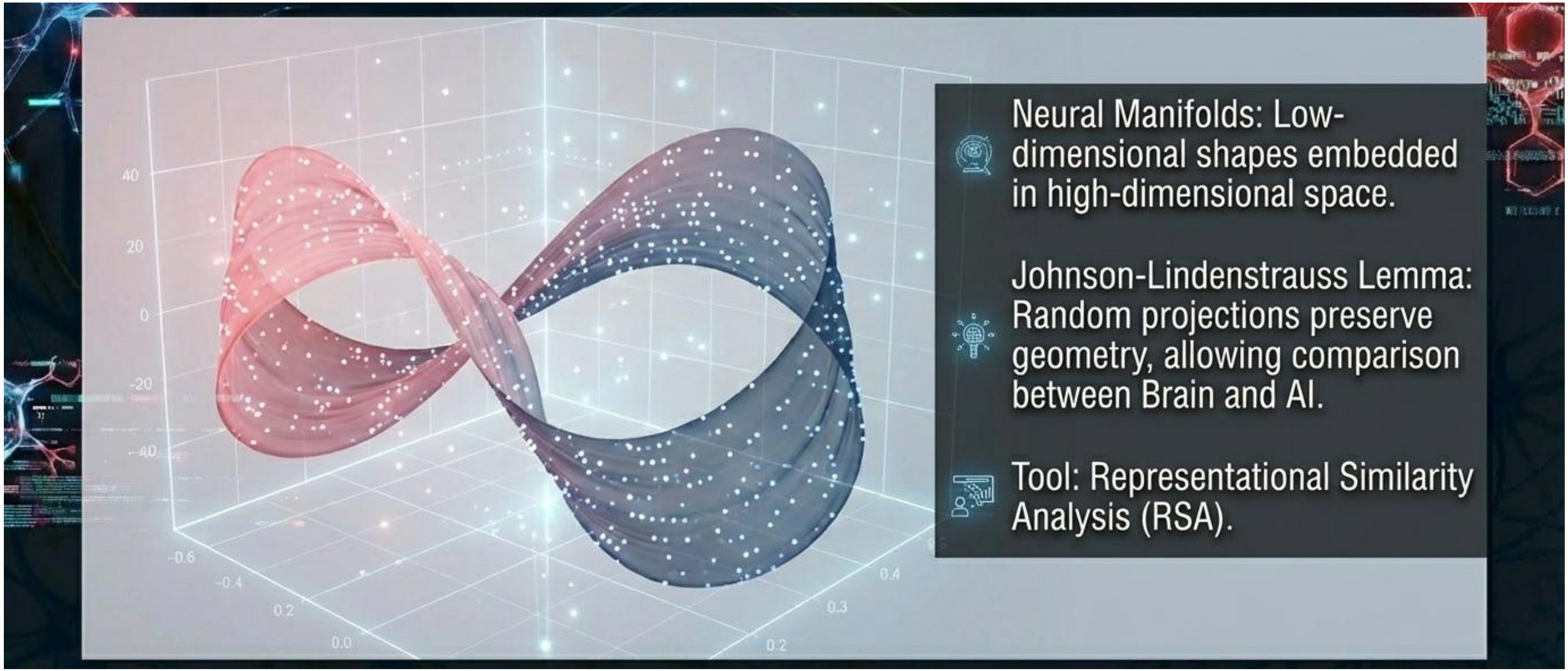
The limitation of Micro View: Polysemanticity



Marco View


Hopfieldian View:
Intelligence emerges from the collective dynamics of a *population*, not single units.
Key Concept: The Neural Population Doctrine.
Analogy: A single pixel tells you nothing; the image exists only in the collective.

Geometry of Thought

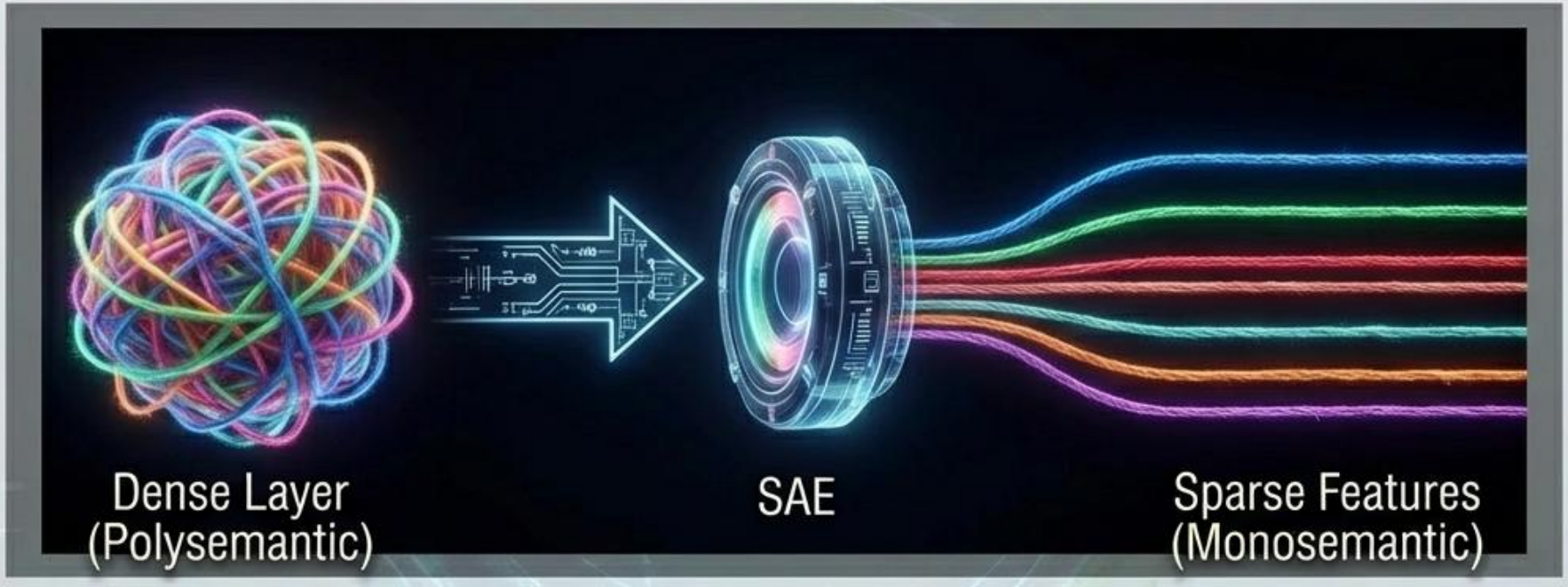


Neural Manifolds: Low-dimensional shapes embedded in high-dimensional space.

Johnson-Lindenstrauss Lemma: Random projections preserve geometry, allowing comparison between Brain and AI.

Tool: Representational Similarity Analysis (RSA).

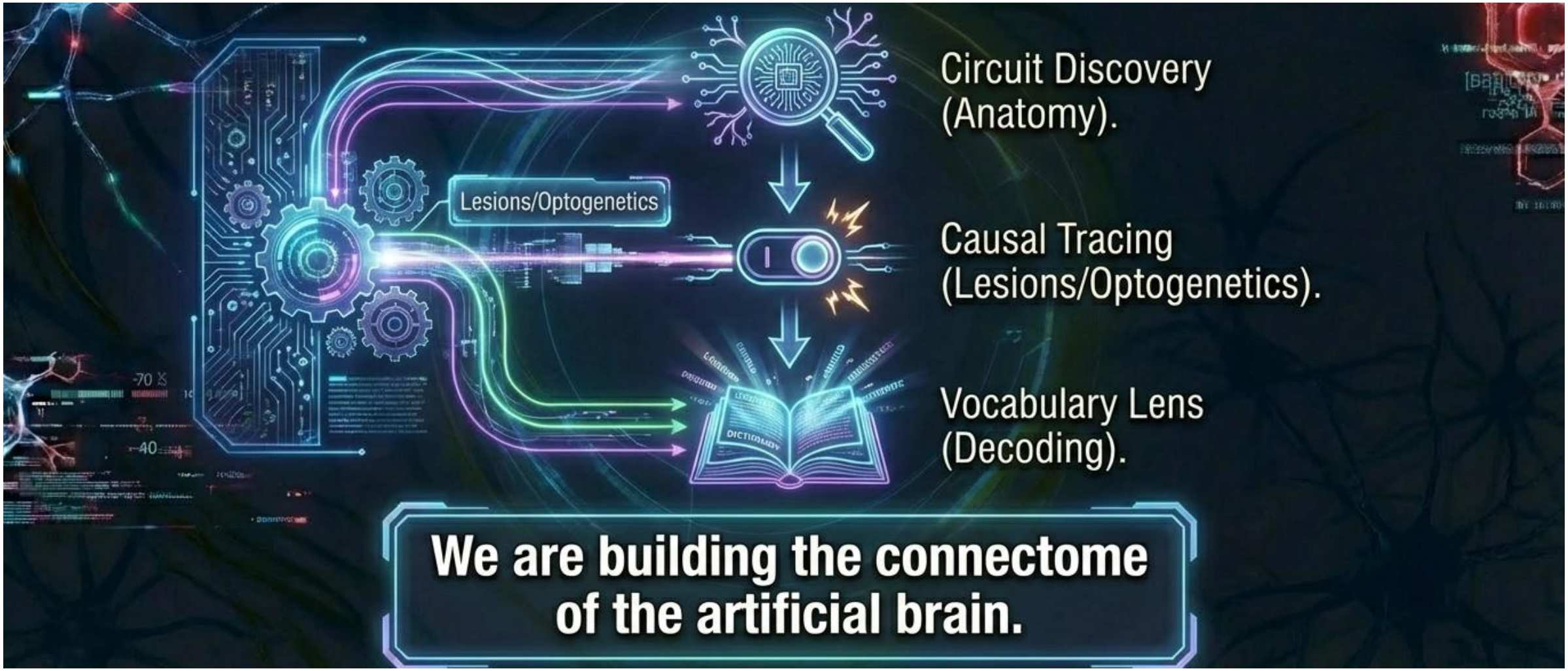
Untangling the Population: Distributed Representations



Mechanism: Decomposing messy population vectors into interpretable feature directions.

Inspiration: Sparse Coding in the V1 Visual Cortex.

Mechanistic Interpretability is Virtual Neuroscience



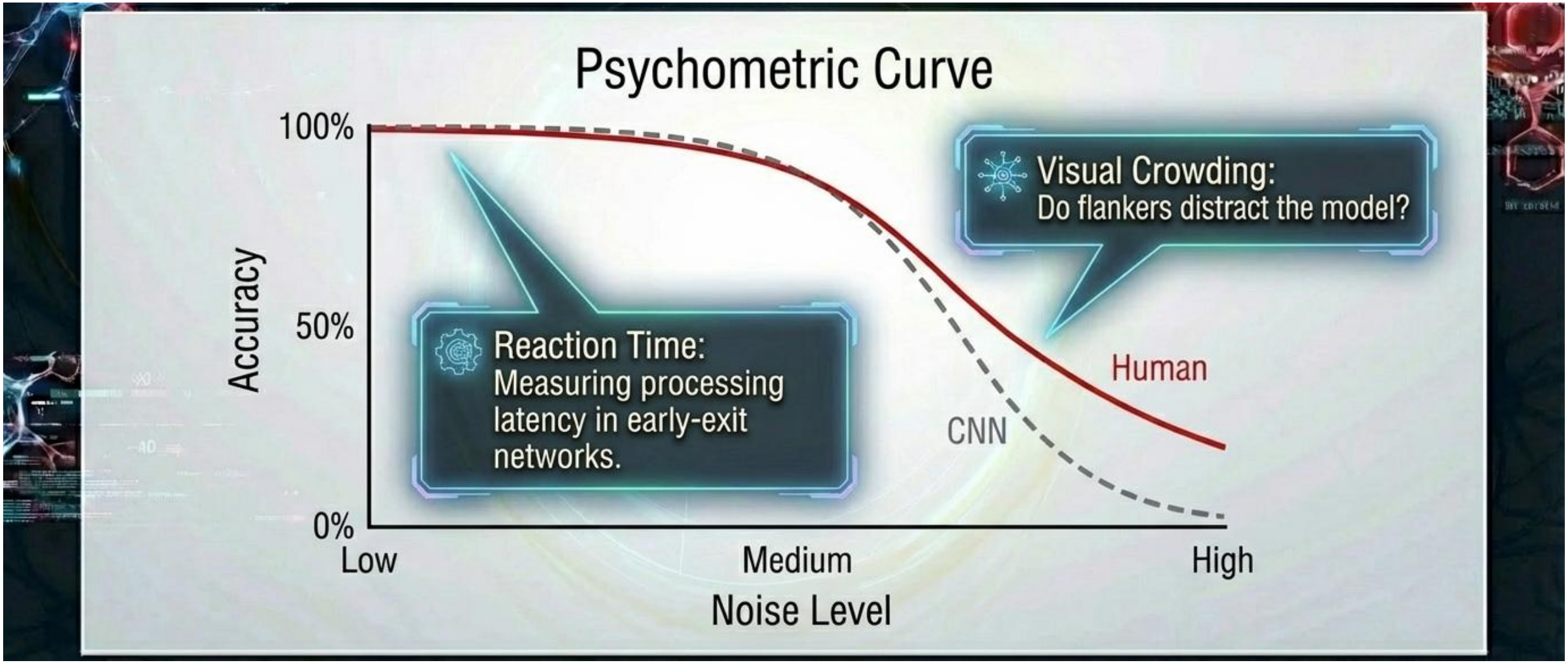
The Behavioural View: Artificial Cognition

Concept: Experimental Psychology for Machines.

Method: Manipulate the Stimulus → Measure the Behavior.

Quote: “The human mind is also a black box… we study it without opening it.”

Psychophysics For Machines



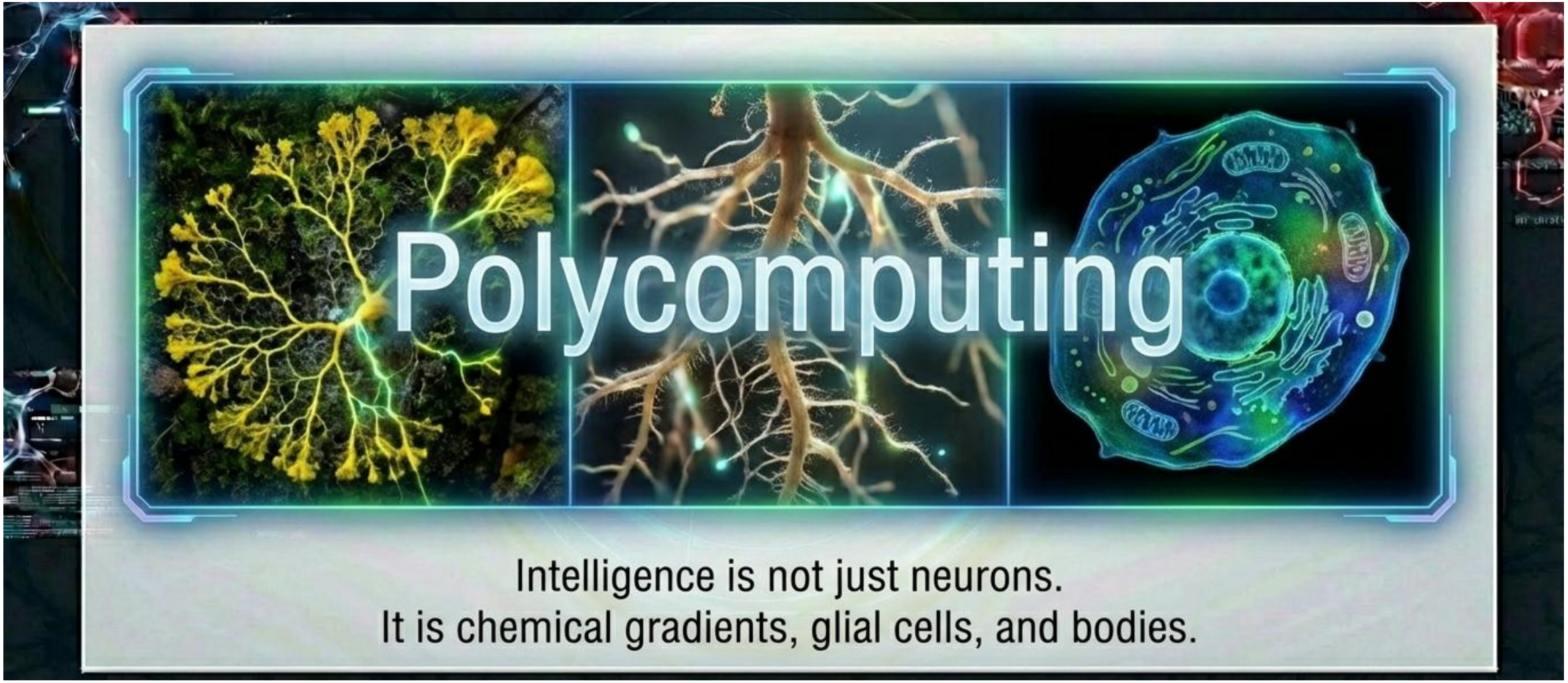
Case Study: The Shape Bias

Human sees:
CAT (Shape Bias).

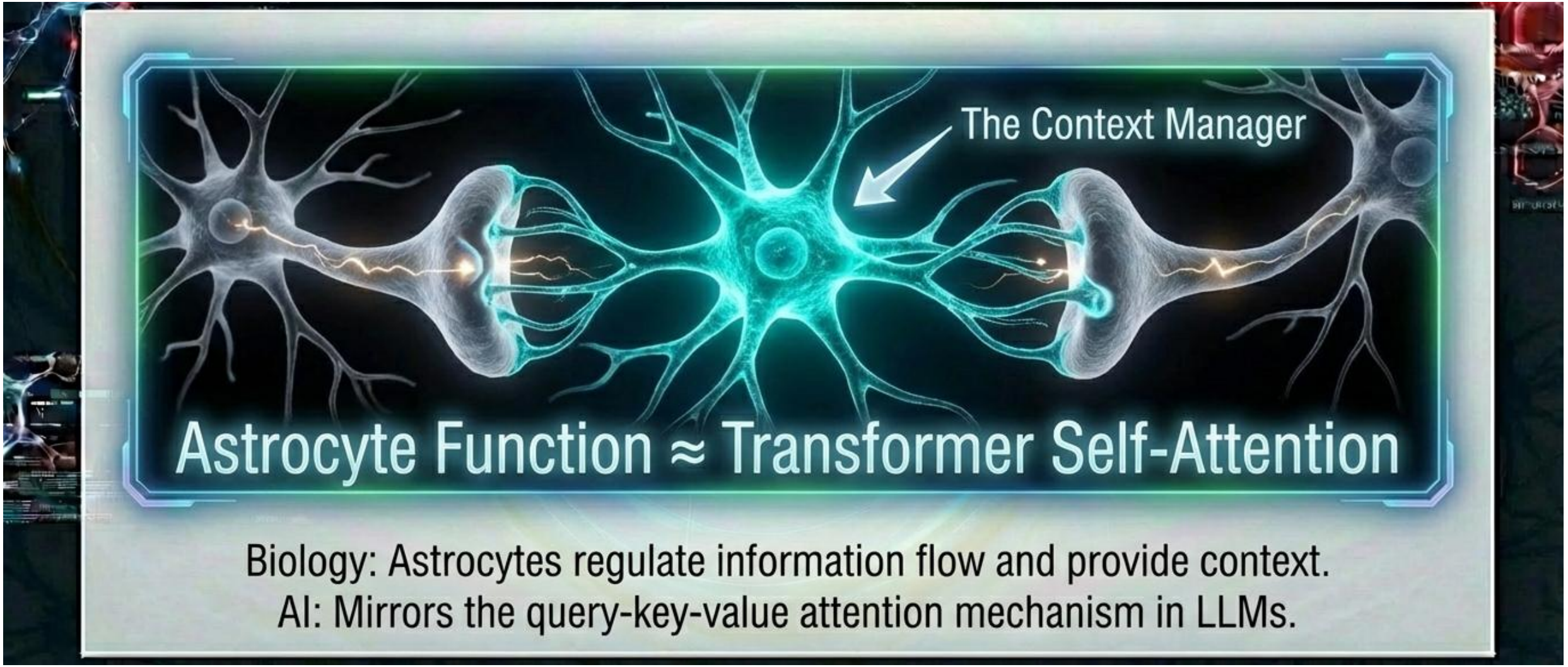
Standard CNN sees:
ELEPHANT (Texture Bias).

Insight: To achieve Human-Like Intelligence, AI must learn to prioritize global geometry over local texture.

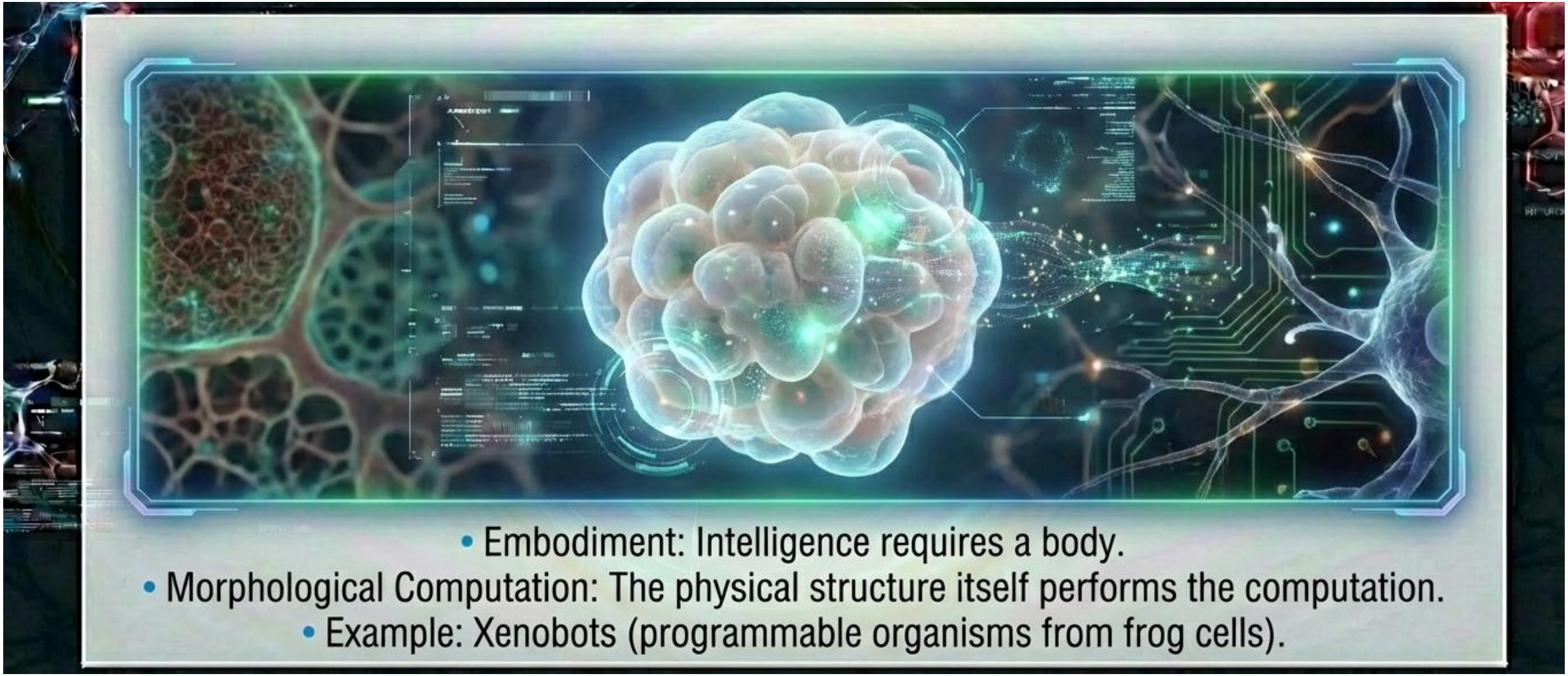
The Architectural View: Bio-Inspired AI



Beyond the Neuron: The Astrocyte

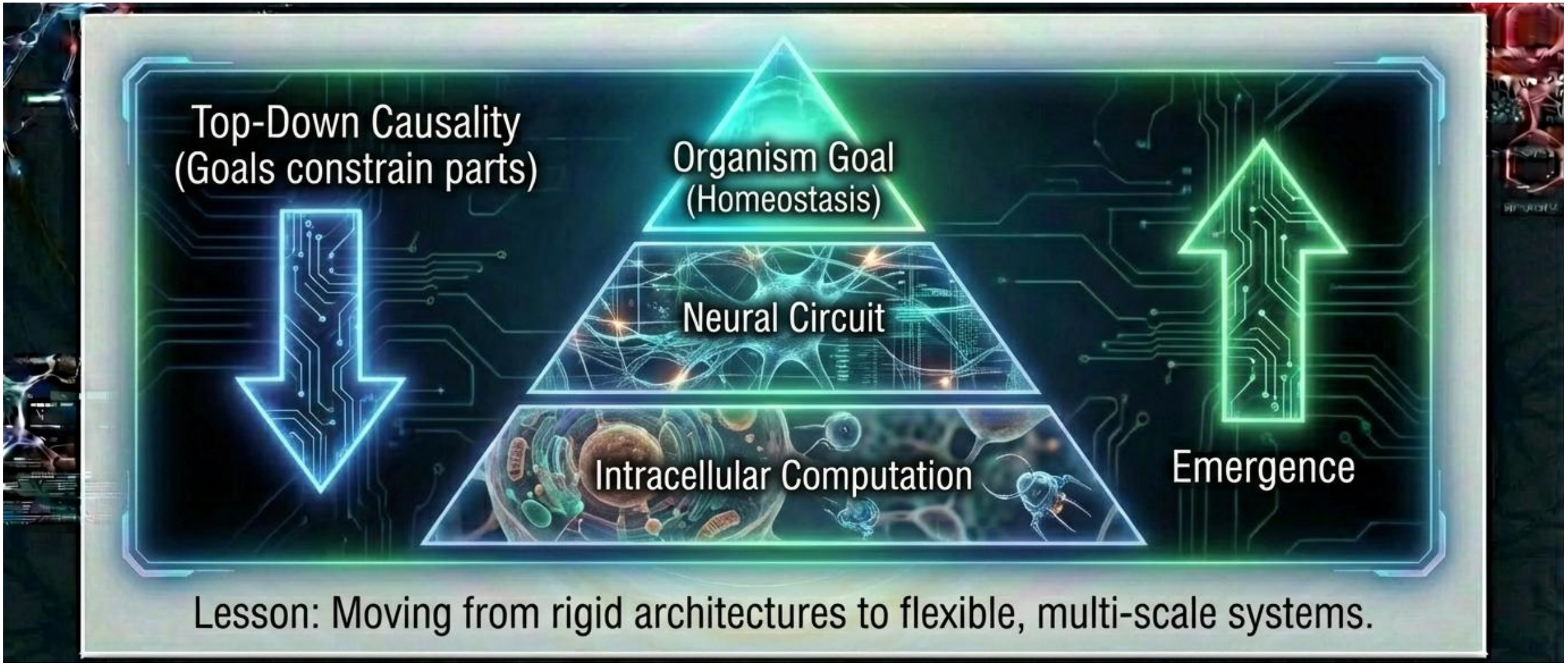


Embodiment & Morphological Computation



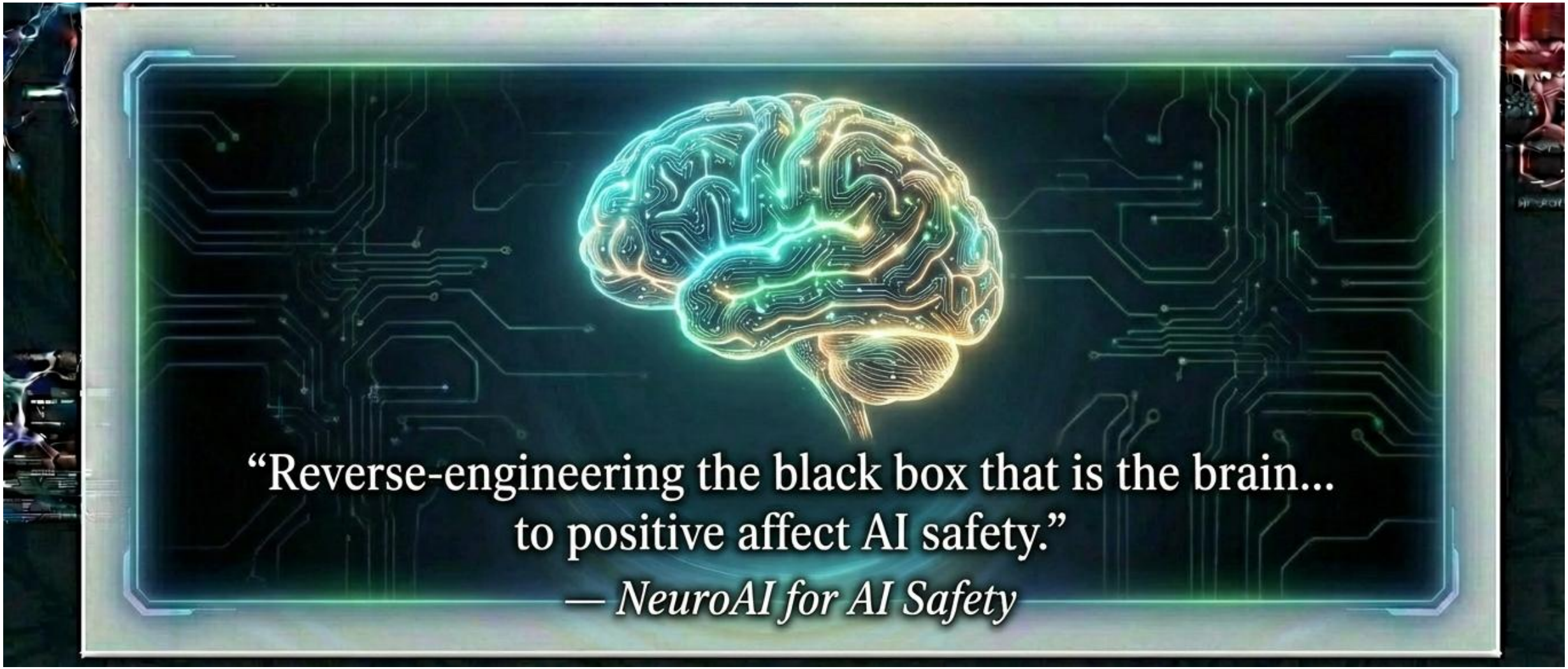
- Embodiment: Intelligence requires a body.
- Morphological Computation: The physical structure itself performs the computation.
- Example: Xenobots (programmable organisms from frog cells).

Hierarchy, Modularity, & Polycomputing



The Roadmap to Human-Like Intelligence

Deciphering the Ghost in the Machine



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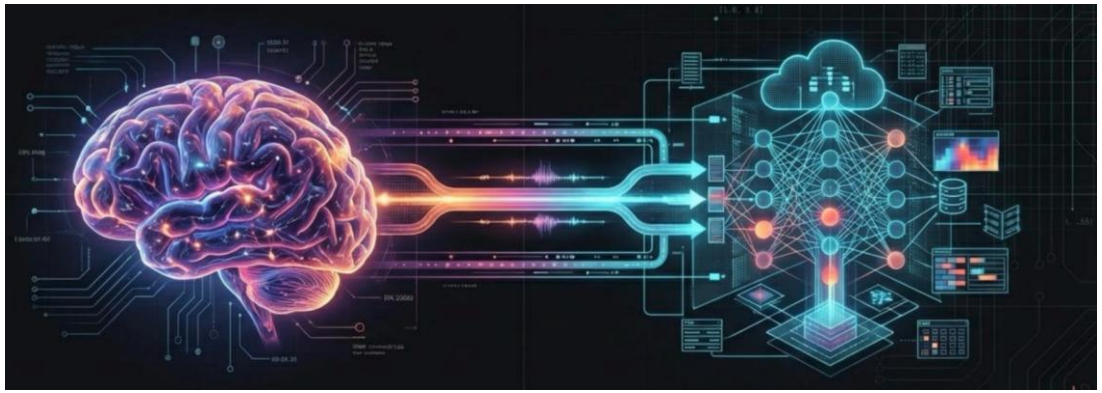
Thanks